

Demand Analysis for Plastic Buckets in the Dairy Industry

DSA 210 Final Report — Omer Dogru — Sabanci University — Spring 2026

1. Motivation

My family owns a plastic injection factory in Osmaniye, Turkey, producing buckets primarily for yogurt manufacturers across the country. Over the years, I noticed that sales tend to increase during warmer months, likely because dairy consumption — and therefore bucket demand — rises with temperature. However, this was just an intuition, never backed by data.

This project aims to systematically analyze the factors affecting weekly bucket demand using real sales data from the factory, combined with external variables such as temperature, oil prices, and exchange rates. As an Industrial Engineering student, understanding demand drivers is directly relevant to operations, inventory management, and capacity planning.

2. Data Sources

The primary dataset consists of weekly plastic bucket sales in tonnage from the family factory, spanning three years (January 2023 to December 2025), totaling 156 weekly observations. The original data was in monthly format and was converted to weekly using a daily-rate weighted method that handles month boundaries naturally.

External datasets collected to enrich the analysis:

- **Temperature and precipitation** — Daily weather for 10 Turkish cities from the Open-Meteo Historical API. A Turkey-wide temperature was computed as a population-weighted average of 9 demand-side cities (Istanbul, Ankara, Izmir, Bursa, Antalya, Konya, Adana, Gaziantep, Kayseri). Osmaniye (factory location) was kept as a separate signal.
- **Brent crude oil price** — Daily Brent futures (BZ=F) from Yahoo Finance, serving as a proxy for plastic raw material costs.
- **USD/TRY exchange rate** — Daily from Yahoo Finance, relevant because plastic raw materials are often imported and priced in USD.

All external data was aggregated to weekly frequency (ISO Monday–Sunday). Financial series were forward-filled across weekends before computing weekly averages. The final merged dataset has 156 rows and 14 columns with zero missing values.

3. Data Analysis

3.1 Exploratory Data Analysis

The sales time series reveals two dominant patterns: a strong upward trend across the three-year period (year-over-year growth of 21.8% in 2024 and 28.0% in 2025), and a recurring seasonal cycle with summer peaks and winter troughs. The year-over-year comparison shows each year sitting above the previous one at virtually every week, suggesting market expansion

independent of seasonal effects.

Scatter plots showed a mild positive relationship between temperature and sales, though with substantial scatter. The Shapiro-Wilk test rejected normality for sales ($p < 0.05$), informing the choice of non-parametric tests in hypothesis testing.

An initially unexpected finding was a strong negative correlation between Brent oil price and sales. The Brent time series shows oil prices declining over 2023–2025 while sales rose steadily. This became a key point of discussion in hypothesis testing.

3.2 Hypothesis Testing

Five hypotheses were tested at $\alpha = 0.05$, using non-parametric methods due to the non-normal distributions observed in EDA:

#	Hypothesis	Test	Statistic	p-value	Result
H1	Summer vs Winter sales differ	Mann-Whitney U	-	0.114	Fail to reject
H2	Temp positively associated with sales	Spearman (one-sided)	$\rho = 0.197$	0.007	Reject H_0
H3	Seasonal sales distributions differ	Kruskal-Wallis	-	0.089	Fail to reject
H4	Brent oil associated with sales	Spearman	$\rho = -0.617$	≈ 0	Reject H_0
H5	USD/TRY change $\leftrightarrow$ sales change	Spearman (1st diff)	$\rho = -0.068$	0.399	Fail to reject

H1 and H3 — Seasonality weaker than expected: While summer sales appear visually higher than winter in boxplots, the Mann-Whitney U test did not reach significance ($p = 0.114$) and Kruskal-Wallis was borderline ($p = 0.089$). The strong year-over-year growth trend creates large within-season variance that masks the seasonal effect. For example, winter 2025 sales are higher than summer 2023 sales simply because the factory grew. A detrended analysis would likely reveal stronger seasonality.

H2 — Temperature has a weak but significant positive effect: Spearman $\rho = 0.197$ with $p = 0.007$ confirms warmer weeks are associated with higher sales, but the effect is modest. The strong growth trend adds noise to the temperature–sales relationship.

H4 — Brent oil shows a strong negative correlation ($\rho = -0.617$): This was the most surprising result. It does not mean cheaper oil causes higher sales; rather, Brent prices declined steadily over 2023–2025 (from ~\$85 to ~\$60) while sales trended upward due to business growth. This is a classic case of spurious correlation between trending time series — the same phenomenon tested in H5.

H5 — No FX–demand link after detrending: Once the common trend is removed via first-differencing, the week-to-week relationship between USD/TRY changes and sales changes disappears ($\rho = -0.068$, $p = 0.399$), confirming the spurious nature of the raw correlation.

3.3 Feature Engineering

Features created for machine learning:

- **Time index** (0 to 155) — captures the overall growth trend, as recommended by the course assistant.
- **Lagged sales** (1-week and 2-week lags) — momentum features.
- **Rolling 4-week average** of lagged sales — smooths weekly noise.
- **Lagged temperature** (1-week lag) — demand may respond with a delay.
- **Season dummies** and **cyclical month encoding** (sin/cos transforms).

After creating lagged and rolling features, 5 rows were dropped due to NaN values, leaving 151 observations for modeling.

3.4 Machine Learning Models

Four regression models were trained using a chronological 80/20 train–test split. No random shuffling was used, as this would leak future information into training.

Model	RMSE	MAE	R^2
Linear Regression	7,330	5,853	0.734
Ridge ($\alpha = 10$)	7,543	6,046	0.718
Random Forest	13,361	10,342	0.115
XGBoost	17,354	13,967	-0.493

Linear Regression achieved the best test performance ($R^2 = 0.734$), explaining about 73% of sales variance on unseen data. Ridge performed similarly ($R^2 = 0.718$), confirming multicollinearity was not a major issue. Tree-based models overfit badly despite constraining depth and leaf size — a known problem with small datasets (~150 observations). An unconstrained Random Forest achieved near-perfect training R^2 but poor test performance, a textbook case of overfitting.

TimeSeriesSplit cross-validation (5 folds) produced negative mean R^2 values for all models. This is expected with strongly trending data: early folds train on 2023 (lower sales) and predict 2024–2025 (higher sales), causing systematic underprediction. This limitation is inherent to trending time series and does not invalidate the single-split results.

3.5 Feature Importance

Random Forest feature importance showed **sales_lag1** (last week's sales) dominating at approximately 80%, followed by **time_index** (~5%) and **usd_try** (~4%). Temperature variables scored lower than expected based on the hypothesis tests.

This apparent contradiction has a logical explanation: **sales_lag1** implicitly carries temperature information. When it is warm, sales are high, so next week's lag is also high. The lag feature absorbs the signal that temperature would otherwise provide. When tested with only temperature as a feature (no lags), model performance dropped substantially, confirming that temperature's effect is real but gets masked in multivariate models.

4. Key Findings

1. **Growth trend dominates the data.** Year-over-year growth of 22–28% annually is the strongest signal. This growth — likely from new customers, capacity expansion, or market share gains — overwhelms seasonal and temperature effects in the raw data.
2. **Temperature has a real but modest effect.** Spearman $\rho = 0.197$ ($p = 0.007$) confirms a positive association between temperature and sales, but the effect is weaker than initially expected, partly because the growth trend adds noise.
3. **Seasonality is present but not statistically significant.** The strong growth trend means winter 2025 sales can exceed summer 2023 sales, inflating within-season variance and weakening standard seasonal tests. A detrended analysis would likely reveal stronger seasonality.
4. **Brent oil correlation is spurious.** The strong negative correlation ($\rho = -0.617$) reflects opposite trends over time (oil declining, sales rising), not a causal link.
5. **Simple models outperform complex ones on small data.** Linear Regression ($R^2 = 0.734$) beat Random Forest and XGBoost, which overfit the 156-week dataset.
6. **Sales momentum is the strongest short-term predictor.** The 1-week lag feature dominated importance at ~80%, making it useful for short-term demand forecasting.

5. Limitations and Future Work

5.1 Limitations

- **Small sample size (156 weeks)** limits complex model performance and statistical power for seasonal tests.
- **Monthly-to-weekly conversion** is an approximation; actual weekly sales data would be more accurate.
- **Growth trend confounds seasonal analysis.** Detrending before hypothesis testing could reveal clearer seasonality.
- **Hidden factors not in the data.** New customer acquisitions, factory capacity changes, competitor closures, and marketing campaigns all affect sales but are not captured in the dataset.
- **Lag-dependent predictions** limit the practical forecast horizon to 1–2 weeks ahead.

5.2 Future Work

- Collect actual weekly sales data instead of converting from monthly.
- Detrend the sales series before seasonal analysis to isolate pure seasonal effects.
- Add dairy sector-specific indicators: milk price index, yogurt production volume from TUIK.
- Include Ramadan period as a binary feature — dairy consumption patterns may shift during fasting.

- Experiment with time series-specific models (ARIMA, Prophet) that handle trends and seasonality natively.
- Incorporate polypropylene price data (the actual raw material) instead of Brent as a cost proxy.
- Collect qualitative data on new customer events and capacity changes to model hidden demand factors.

References

- [1] Open-Meteo Historical Weather API — <https://open-meteo.com/>
- [2] Yahoo Finance (Brent Crude, USD/TRY) — via yfinance Python library
- [3] scikit-learn: Machine Learning in Python — Pedregosa et al. (2011)
- [4] XGBoost: A Scalable Tree Boosting System — Chen & Guestrin (2016)